Unit-02: Random variables and probability distributions

1. A hole is drilled in a metal sheet component, and then a shaft is inserted through the hole. The shaft clearance is equal to the difference between the radius of the hole and the radius of the shaft. Let the random variable X denote the clearance, in millimeters. The probability density function of X is defined by :

$$f(x) = \begin{cases} 1.25(1 - x^4) & 0 < x < 1 \\ 0 & otherwise \end{cases}$$

What is the probability that 0.8 mm of the components are scrapped? Find also the CDF of the random variable X.

Ans: 0.0819

2. The lifetime in months of a transistor in a certain application is random with probability density function given by

$$f(x) = \begin{cases} 0.1\, e^{-0.1t} & t \geq 0 \\ 0 & t \leq 0 \end{cases}$$

a) Find the mean lifetime. **Ans:** $\frac{\mathbf{100}}{(\ln\{e\})^2}$

b) Find the standard deviation of the lifetimes. **Ans:** $\frac{\mathbf{2000}}{(\ln\{e\})^2} - \left(\frac{\mathbf{100}}{(\ln\{e\})^2}\right)^2$

d) Find the probability that the lifetime will be less than 12 months. **Ans:** $\mathbf{1 - e^{-1.2}}$

3. In a medical examination the chances of error are 15%. Find the Bernoulli distribution if one patient is randomly selected out of 60 patients.

Ans: $P(X = 1) = \frac{51}{60} = 0.85$ and $P(X = 0) = \frac{9}{60} = \mathbf{0.15}$

4. The probability of a shooter hitting a target is $1/3$. How many times he should shoot so that the probability of hitting the target at least once is more than ¾.
   Ans: The shooter must shoot at least 4 times.

5. If the mean and standard deviation of the number of correctly answered question in a test given to 4096 students is 2.5 and $\sqrt{1.875}$. Find an estimate of the number of candidates answering correctly
   - **(i)** 8 or more questions — **Ans: 2 candidates answer correctly 8 or more questions.**
   - **(ii)** 2 or less questions — **Ans: 2153 candidates answer correctly 2 or less questions.**
   - **(iii)** 5 questions. — **Ans: 239 candidates answer correctly 5 questions.**

6. In a quiz contest of answering 'yes' or 'no', what is the probability of guessing at least 6 answers correctly out of ten question asked? Also find the probability of the same if there are 4 options for a correct answer.

   Ans: $\boldsymbol{P(x \geq 6) = P(6)+\ldots+P(10) = 0.377}$ **with two options for a correct answer.**

   $\boldsymbol{P(x \geq 6) = P(6)+\ldots+P(10) = 0.019}$ **with four options for a correct answer.**

7. A shop has 4 diesel generator sets which it hires every day. The demand for generator set on an average is a Poisson variate with value 5/2.Obtain the probability that on a particular day (i) there was no demand (ii) a demand has to be refused.

   Ans : (i)0.0820 (ii) 0.1088

8. Assume that the probability of an individual coal miner being killed in a mine accident during a year is 1/2400. Calculate the probability that in a mine employing 200 miners, there will be at least one fatal accident in a year.
   Ans: 0.08

9. Suppose that a book of 600 pages contain 40 printing mistakes. Assume that these errors are randomly distributed throughout the book and the number of errors per page has a Poisson distribution. What is the probability that 10 pages selected at random will be free of errors?

   Ans: 0.51

10. In an examination taken by 500 candidates, the average and the standard deviation of marks obtained (normally distributed) are 40% and 10%. Find approximately
    (i) how many will pass, if 50% is fixed as a minimum?
    Ans: The required number of candidates is equal to 79.
    (ii) what should be the minimum if 350 candidates are to pass?
    Ans: 35% minimum pass marks could enable 350 candidates to pass out of 500 candidates.
    (iii) how many have scored marks above 60%?
    Ans: The required number of candidates who scored more than 60% marks = $500 \cdot 0.0228 \approx 11$ (approximately).
11. The mean inside diameter of a sample of 200 washers produced by a machine is 5.02mm and the standard deviation is 0.05mm. The purpose for which these washers are intended allows a maximum tolerance in the diameter of 4.96 to 5.08mm, otherwise the washers are considered defective. Determine the percentage of defective washers produced by the machine, assuming the diameters are normally distributed.

    Ans: 23.02%

12. In a test on 2000 electric bulbs, it was found that the life of a particular make, was normally distributed with an average life of 2040 hours and S.D. of 60 hours. Estimate the number of bulbs likely to burn for
    (a) more than 2150 hours,

    (b) less than 1950 hours,

    (c) more than 1920 hours and but less than 2160 hours.

Ans:

(a) The number of bulbs likely to bum for more than 2150 hours is $2000 \times 0.0336 = 67.2 \sim 67$.
(b) The number of bulbs likely to bum for less than 1950 hours is $2000 \times 0.0668 = 133.6 \sim 134$.
(c) The number of bulbs likely to bum for more than 1920 hours but less than 2100 hours is $2000 \times 0.8185 = 1637$.